

Analysis of Seasonal Disease Patterns Using Government Health Data

From Government Records to Seasonal Patterns and Local Field Context in Maharashtra

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PROJECT GUIDE

**[Confirm before
submission]**

SURVEILLANCE PERIOD

2022-2026 W32
N=237 Weeks

ACT 01

What is the Question?

When do diseases appear? And do their patterns repeat across years?

WHEN

does outbreak
activity
concentrate?

WHICH

diseases show
recurring
patterns?

WHERE

are events
geographically
localized?

HOW STABLE

are patterns under
statistical
testing?

Where did the data come from?

Source: National Centre for Disease Control (NCDC) / Integrated Disease Surveillance Programme (IDSP)

237

SURVEILLANCE WEEKS

203 Outbreak • 34 NIL Weeks

809

ANALYTICAL RECORDS

Verified Maharashtra Rows

21,955

REPORTED CASES

2022-2025 Primary Baseline

Extraction & Taxonomic Discipline

DATA CLEANING DISCIPLINE · RIGOROUS REDUCTION

116 Raw Strings → 5 Primary Analytical Cohorts

How unstructured government text becomes statistically testable cohorts without discarding records or inventing categories.

STAGE 01

116

RAW STRINGS

Chaotic OCR variants, abbreviations, mixed casing, and clinical spellings in weekly PDF bulletins.

Dengue Dengue Fever DENGUE

STAGE 02

56

CLEAN LABELS

Harmonized canonical disease labels created via deterministic dictionary lookup while preserving raw provenance.

Dengue Acute Diarrheal Disease Malaria

STAGE 03

30

SYNDROMIC FAMILIES

Etiological and transmission groupings (Vector-borne, Enteric, Zoonotic, Respiratory, Food-borne).

Vector-Borne Arboviral Water-Borne Enteric
Vector-Borne Protozoal

STAGE 04

93.9% RECORDS

5

PRIMARY COHORTS

High-burden analytical cohort capturing 93.9% of all 2022-2025 baseline records (539 of 574 records).

Dengue (204) ADD (153) Malaria (70)

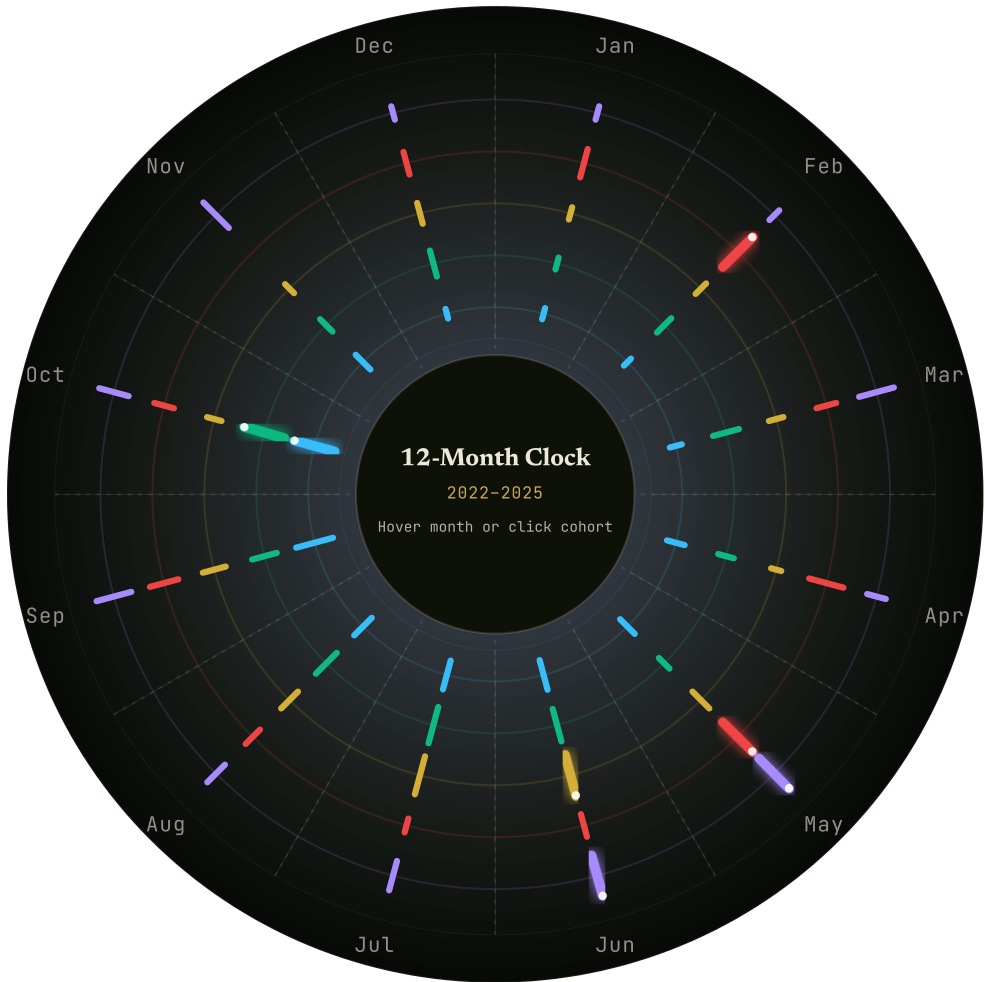
INSPECTION: STAGE 04 (PRIMARY COHORTS)

Isolated 5 core cohort families representing 21,955 cases and 266 deaths.

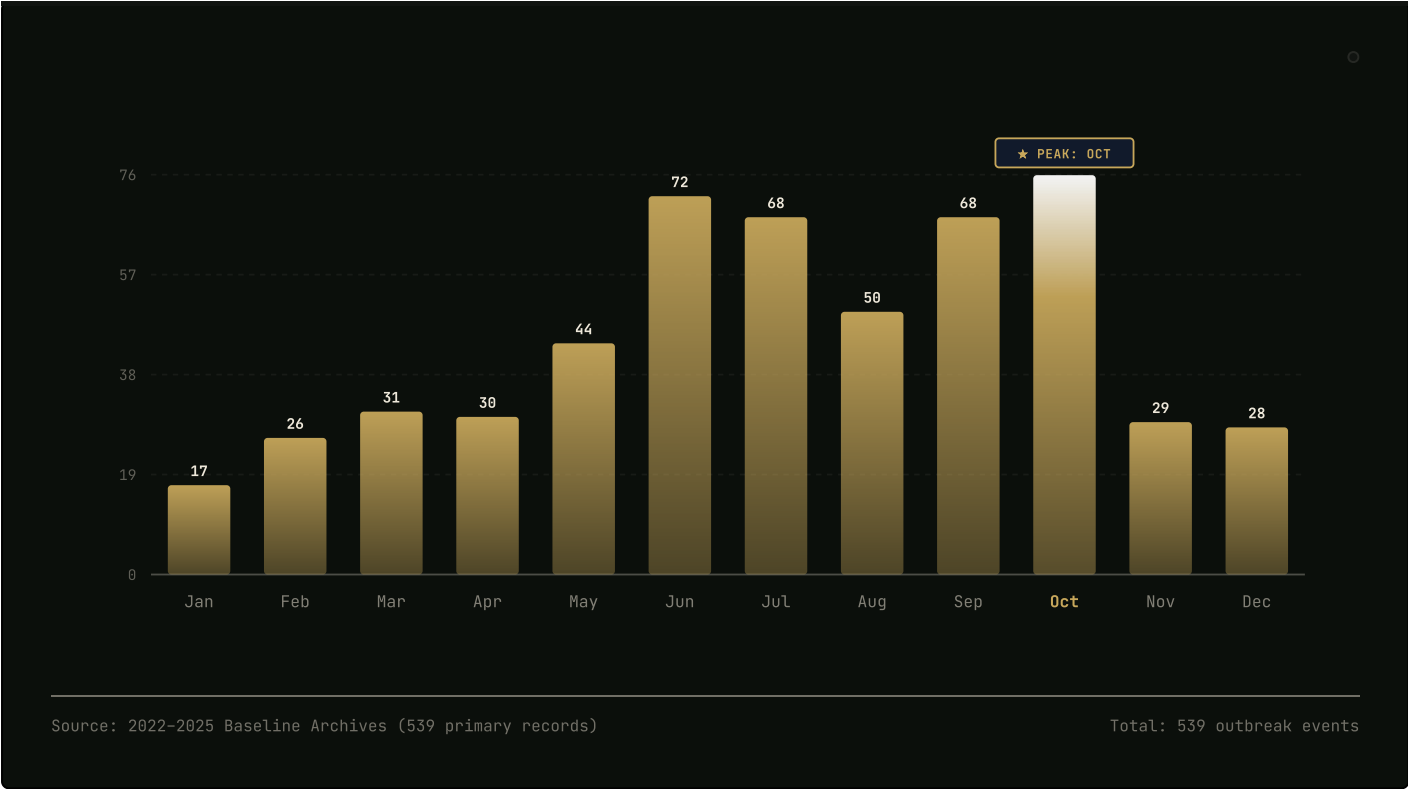
Additive normalization: 'disease_raw' string permanently preserved in database.

When do these diseases appear?

Five Diseases · Five Temporal Signatures



Concentric seasonal rings reveal distinct temporal niches: Malaria peaks early (May-Jul), ADD spans the monsoon, Dengue rises late (Aug-Oct), and Food Poisoning is non-monsoonal.

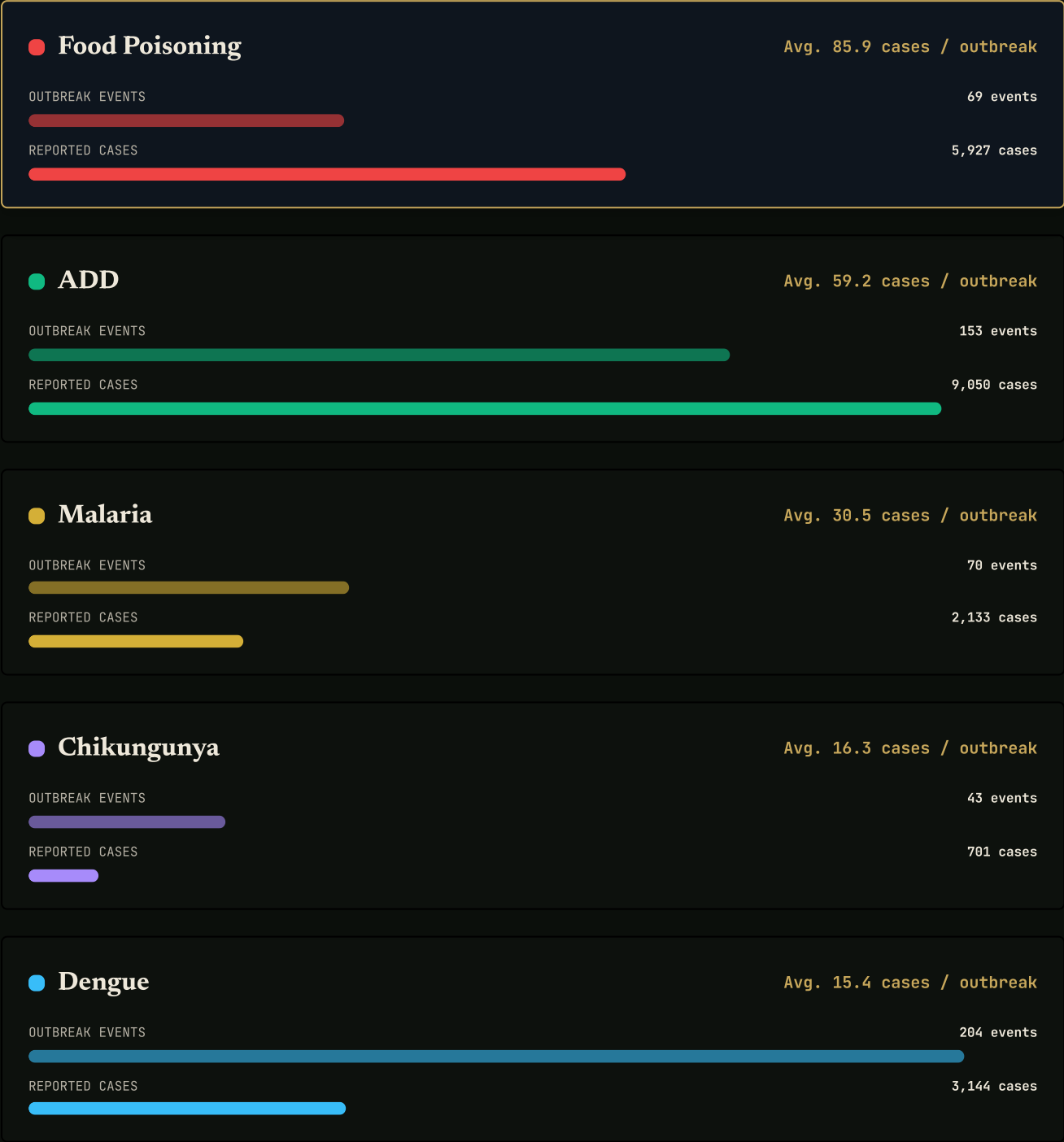


Concordance & Case Divergence

ANALYTICAL CONTRAST · 2022-2025 BASELINE

Are more events always more cases?

Outbreak event count measures **how often** clusters are reported. Case count measures **how many people** were affected.



FOCUS ANALYSIS: FOOD POISONING

High cluster severity: large wedding/feast mass-exposure events skew case volume.

69

EVENTS

5,927

CASES

85.9

CASES / EVENT

EXPLORATORY MONTHLY CONCORDANCE • 2022-2025

Do the monthly ranks repeat each year?

Kendall's W evaluates how consistently the 12 calendar months maintain the same relative rankings across completed baseline years.

★ STATISTICALLY SIGNIFICANT (P < 0.05)

HIGHEST-ACTIVITY MONTHS ACROSS 4 BASELINE YEARS

Year 2022

Rank #1	Jul
Rank #2	Jun
Rank #3	Aug
Rank #4	Oct

Year 2023

Rank #1	Jul
Rank #2	Aug
Rank #3	Jun
Rank #4	Oct

Year 2024

Rank #1	Jun
Rank #2	Jul
Rank #3	Aug
Rank #4	Oct

Year 2025

Rank #1	Jul
Rank #2	Aug
Rank #3	Oct
Rank #4	Jun

CONCORDANCE W

0.530

P-VALUE

0.016

Statistically significant rank concordance (p = 0.016). The 12 calendar months maintain consistent relative rankings across all 4 baseline years, driven by repeated monsoon-onset contamination.

ACT 08

Geographic Footprint: Maharashtra

Statewide seasonal synchronization does not imply geographic uniformity.

DISTRICT REPORTING FOOTPRINT

31 districts reporting ·



Dengue: 31 Districts
Widespread post-monsoon footprint

ADD: 33 Districts
Broadest reporting breadth

Malaria: 61.4% in 2 Districts
Gadchiroli & Chandrapur cluster

Outlier Sensitivity Decomposition

SENSITIVITY ANALYSIS • POINT-SOURCE CLUSTER IMPACT

Food Poisoning: Can 3 records alter the calendar peak?

In February, 3 single outbreak events account for 1,615 of 2,162 reported cases (74.7%).

2024 • W07

SINGLE EVENT

Kolhapur

651

REPORTED CASES

2025 • W06

SINGLE EVENT

Parbhani

629

REPORTED CASES

2023 • W06

SINGLE EVENT

Solapur

335

REPORTED CASES

FEBRUARY CASE VOLUME

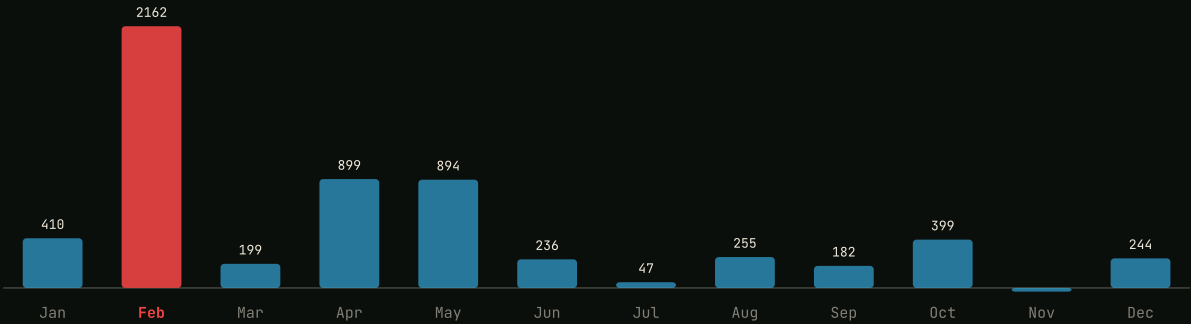
2,162 cases

Complete raw baseline total

PRIMARY PEAK TIMING

February (2,162 cases)

Dominated by single-event wedding clusters



METHODOLOGICAL NOTE: These three records were **never deleted** from the research database. Sensitivity analysis decomposes aggregate statistics to show how much point-source exposure clusters influence calendar peaks.

2026 Partial-Year Surveillance Window

SURVEILLANCE BOUNDARY • OUT-OF-SAMPLE WINDOW

2026 Partial Window: W01 through W32

Surveillance observation ends strictly at Week 32. It cannot be used as an annual baseline or full-year forecast.



45

PRIMARY COHORT RECORDS

Across 5 disease families

1,118

REPORTED CASES (W01-W32)

6 Dengue • 15 ADD • 19 FP • 4 Mal • 1 Chik

0

REPORTED DEATHS

Zero fatalities recorded in window

TRUNCATION WARNING

Historically, 55.4% of all Dengue outbreak events occur after Week 32. Evaluating 2026 Dengue without post-monsoon weeks produces a false impression of suppression.

TAXONOMIC CLASSIFICATION IN 2026

All 19 Food Poisoning events recorded in 2026 W01-W32 were explicitly categorized as "Suspected" in the NCDC PDF bulletins.

Fieldwork Evidence & Contextual Comparison

FACILITY VERIFICATION

PHC Nallasopara West – VVMC Health Post

Nallasopara West, Kala Krida Ground area, Palghar, Maharashtra • 02 Oct 2026, 10:30-11:45 AM

One local facility • One structured environmental observation • One Medical Officer In-Charge key-informant interview.

GOVERNMENT DATA (NCDC)

- Dengue: Jun-Oct concentration (Peak Oct)
- Malaria: May-Jul surge (Vidarbha cluster)
- ADD: Jun-Aug peak (Kendall $W=0.530$, $p=0.016$)
- Food Poisoning: Bimodal Jan-Feb & Apr-May
- Chikungunya: No stable recurring pattern

LOCAL MEDICAL OFFICER REPORTED

- Dengue: Sep-Nov clinical peak (drum storage)
- Malaria: Jul-Sep (monsoon construction pools)
- ADD: Jun-Aug onset surge (drain overflow)
- Food Poisoning: Sudden cohort arrivals after feasts
- Chikungunya: Sporadic household clusters

“The NCDC/IDSP layer summarizes reported surveillance events across Maharashtra, while the field interview represents one local government health facility. The comparison is contextual rather than a statistical validation exercise.”

ACT 14

Conclusion

The pattern is seasonal. The context is local. The interpretation needs both.